

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Open-label Study Comparing Efficacy and Safety of Sacituzumab Tirumotecan (Sac-TMT, MK-2870) as a Monotherapy and in Combination With Pembrolizumab (MK-3475) Versus Treatment of Physician's Choice in Participants With Previously Untreated Locally Recurrent Unresectable or Metastatic Triple-Negative Breast Cancer Expressing PD-L1 at CPS Less Than 10 (TroFuse-011) **Short Title/ Acronym:** TroFuse-011 / MK-2870-011 **Protocol Number:** MK-2870-011 **NCT Number:** NCT06841354 **Trial Status:** Recruiting **Sponsor Name:** Merck Sharp & Dohme LLC (MSD) / Merck **Investigational Product:** Sacituzumab Tirumotecan (Sac-TMT, MK-2870) and Pembrolizumab (MK-3475) **Phase of Development:** Phase III **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To learn if participants treated with sacituzumab tirumotecan alone or in combination with pembrolizumab live longer overall (Overall Survival) or without the cancer growing or spreading (Progression-Free Survival) compared to participants treated with standard chemotherapy.

Secondary Objectives: To evaluate Objective Response Rate (ORR), Duration of Response (DOR), Patient-Reported Outcomes (PROs), overall safety, and additional Progression-Free Survival (PFS) and Overall Survival (OS) comparisons between the investigational arms.

2.2 Background & Rationale The standard first-line therapy for most previously untreated locally recurrent unresectable or metastatic Triple-Negative Breast Cancer (TNBC) with a PD-L1 Combined Positive Score (CPS) < 10 is chemotherapy alone. There is a significant medical necessity to establish new therapeutic options with superior efficacy for this population. Sacituzumab tirumotecan (Sac-TMT) is a novel antibody-drug conjugate (ADC) targeting TROP2. There is strong biologic rationale supporting the combination of this ADC with the immunotherapy pembrolizumab (anti-PD-1) to improve therapeutic outcomes regardless of high PD-L1 expression.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Treatment of Physician's Choice [TPC]: paclitaxel, nab-paclitaxel, or gemcitabine + carboplatin)

Blinding: Open-label **Randomization:** 2:1:2 (Arm A [Sac-TMT] : Arm B [Sac-TMT + Pembrolizumab] : Arm C [TPC])

3.2 Planned Enrollment & Duration Target **\$N\$:** 1000 participants
Number of Sites: Global multicenter trial **Estimated Study Start:** 16-Mar-2025 **Estimated Primary Completion:** 18-May-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has locally recurrent unresectable or metastatic TNBC that cannot be treated with curative intent. 2. Has not received systemic treatment for locally recurrent unresectable or metastatic breast cancer. 3. Participants previously treated for early-stage breast cancer must have completed all prior therapy for early-stage breast cancer with curative intent at least 6 months before the first disease recurrence. 4. Is a candidate for treatment with pembrolizumab and one of the TPC options: paclitaxel, nab-paclitaxel, or gemcitabine + carboplatin. 5. Participants who have AEs due to previous anticancer therapies must have recovered to $\leq$ Grade 1 or baseline with the exception of alopecia or vitiligo. Participants with endocrine-related AEs who are adequately treated with hormone replacement are eligible. 6. Participants who are hepatitis B surface antigen (HBsAg) positive are eligible if they have received hepatitis B virus (HBV) antiviral therapy for at least 4 weeks, and have undetectable HBV viral load. 7. Participants with a history of hepatitis C virus (HCV) infection are eligible if HCV viral load is undetectable.

4.2 Exclusion Criteria 1. Has breast cancer amenable to treatment with curative intent. 2. Has TNBC with evaluable tumor programmed death ligand 1 (PD-L1) expression at combined positive score (CPS) ≥ 10 . 3. Has received prior systemic therapy for the treatment of locally recurrent unresectable or metastatic breast cancer. 4. Has Grade ≥ 2 peripheral neuropathy. 5. Has a history of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease that prevents/delays corneal healing. 6. Has active inflammatory bowel disease requiring immunosuppressive medication or a previous history of inflammatory bowel disease. 7. Has uncontrolled, significant cardiovascular disease or cerebrovascular disease. 8. Has skin-only metastatic disease. 9. Has advanced/metastatic, symptomatic visceral spread at risk of rapidly evolving into life-threatening complications. 10. Human immunodeficiency virus (HIV)-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 11. Has known additional malignancy that is progressing or has required active treatment within the past 5 years. 12. Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis. Participants with previously treated brain metastases may participate provided they are radiologically stable. 13. Active autoimmune disease that has required systemic treatment in the

past 2 years. Replacement therapy (e.g., thyroxine, insulin, or physiologic corticosteroid) is allowed. 14. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 15. Concurrent active Hepatitis B (defined as HBsAg positive and/or detectable HBV DNA) and Hepatitis C virus (HCV) (defined as anti-HCV Ab positive and detectable HCV RNA) infection. 16. History of stem cell/solid organ transplant. 17. Has not adequately recovered from major surgery or has ongoing surgical complications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * Arm A:

Sacituzumab tirumotecan 4 mg/kg given every 2 weeks (Q2W). * Arm B: Sacituzumab tirumotecan 4 mg/kg Q2W PLUS pembrolizumab 400 mg every 6 weeks (Q6W). * Arm C (Control): Physician's choice of chemotherapy (e.g., paclitaxel Q1W or Q4W, nab-paclitaxel Q4W, or gemcitabine + carboplatin Q3W). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Until disease progression or unacceptable toxicity. Pembrolizumab will be administered for up to 18 doses (approximately 2 years).

5.2 Concomitant & Prohibited Therapy Permitted: Rescue medications per approved product label at the discretion of the investigator; standard supportive care for infusion reactions. **Prohibited:** Concurrent investigational agents or additional systemic anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy (PD-L1/TROP2 testing), Baseline Scans
Baseline	Day 0	Randomization (2:1:2), First Dose of Study Intervention
Interim	Every 2 to 6 Weeks	Intravenous Infusions, CT/MRI Scans, Safety Labs (Blood/Urine), ECGs
End of Study	30 Days Post-Treatment	Final Safety Assessment, Long-term follow-up for Overall Survival

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Progression-Free Survival (PFS):** Assessed comparing sac-TMT versus Treatment of Physician's Choice (TPC); and sac-TMT plus pembrolizumab versus TPC. PFS is defined as the time from randomization to the first documented progressive disease (PD) per RECIST 1.1 based on Blinded Independent Central Review (BICR) or death due to any cause. * **Overall Survival (OS):** Assessed comparing sac-TMT versus TPC. OS is defined as the time from randomization to death due to any cause.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS):

Comparing sac-TMT plus pembrolizumab versus TPC; and sac-TMT plus pembrolizumab versus sac-TMT. * **Progression-Free Survival (PFS):** Comparing sac-TMT plus pembrolizumab versus sac-TMT. * **Objective Response Rate (ORR):** Comparing sac-TMT versus TPC; and sac-TMT plus pembrolizumab versus TPC. * Duration of Response (DOR), Patient-Reported Outcomes (PROs), and overall safety.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Blood and urine testing, physical exams, and Electrocardiograms (ECGs) monitored continuously during the treatment phase and for 30 days after the last dose. Evaluated using CTCAE criteria. **Serious Adverse Events (SAEs):** Reported within standard regulatory timelines (typically 24 hours). **Data Monitoring Committee (DMC):** External independent data monitoring committee to routinely evaluate safety and efficacy signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints (PFS and OS); Safety Analysis Set for all patients receiving at least one dose of the study drug. **Sample Size Justification:** 1000 participants randomized in a 2:1:2 ratio to provide adequate statistical power to detect significant differences in PFS and OS across treatment arms. **Handling of Missing Data:** Standard survival analysis methods (e.g., Kaplan-Meier estimation, censoring at last known alive/progression-free date).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular on-site and remote monitoring visits to ensure protocol adherence and safety. **Audit Trail:** Electronic Case Report Form (eCRF) tracking with independent BICR for radiographic tumor assessments.

11. Study Identifiers & Governance

Primary ID: MK-2870-011 **Registry Number:** NCT06841354 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** TroFuse-011 **Official Regulatory Title:** A Phase 3, Randomized, Open-label Study Comparing Efficacy and Safety of Sacituzumab Tirumotecan (Sac-TMT, MK-2870) as a Monotherapy and in Combination With Pembrolizumab (MK-3475) Versus Treatment of Physician's Choice in

Participants With Previously Untreated Locally Recurrent
Unresectable or Metastatic Triple-Negative Breast Cancer
Expressing PD-L1 at CPS Less Than 10 (TroFuse-011)

12. Clinical Framework (TNBC Specific)

Primary Condition: Triple Negative Breast Neoplasms (TNBC)
Diagnosis Threshold: Locally recurrent unresectable or metastatic TNBC with PD-L1 CPS < 10 **Medical Methodology:** TROP2-targeted Antibody-Drug Conjugate (ADC) therapy $\pm$ PD-1 Immunotherapy
Optimization Target: Progression-Free Survival (PFS) and Overall Survival (OS) maximization

13. Study Procedures & Methodology

Management Pathway: Standard oncology pathway for metastatic breast cancer treatment and monitoring. **Education Protocol (HCP):** Protocol training, RECIST 1.1 evaluation training, and adverse event management for ADCs. **Education Protocol (Patient):** Informed consent, infusion schedule management, and Patient-Reported Outcomes (PRO) questionnaire training. **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for imaging and independent data safety monitoring.

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 2 years of active therapy (for pembrolizumab), or until disease progression, followed by long-term OS follow-up through May 2030. **Onsite Visit Cadence:** Every 2 to 6 weeks correlating with the IV infusion cycles. **Remote Monitoring Frequency:** Routine pharmacovigilance and interval survival checks post-progression. **Checklist Review Interval:** Regular interval scans (CT/MRI) per protocol to monitor tumor response.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				